Opportunity Radar — Testing & Validation Coverage Report

System: TradeBuddy / Opportunity Radar
Market: NSE F&O
Validation Date: 29 September 2026

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1. Executive Overview

The Opportunity Radar was tested across a broad set of market conditions, strategy-selection scenarios, option-chain conditions, risk conditions, data-quality failures, timing conditions and deliberately incorrect trade constructions.

The objective was not simply to check whether the application runs successfully. The testing was designed to determine whether the Radar:

- Detects valid market opportunities.
- Rejects invalid or unsafe opportunities.
- Selects strategies according to market conditions.
- Handles conflicting market signals.
- Respects liquidity requirements.
- Validates option-chain availability.
- Calculates trade risk correctly.
- Prevents structurally invalid trades.
- Handles poor or corrupted market data safely.
- Handles expiry and market-session conditions correctly.
- Produces deterministic decisions for identical inputs.
- Identifies intentionally introduced errors.

The technical validation report records 3,234 automated tests, 607 market scenarios across four contract profiles, 1,097 independently re-verified suggested trades, and 17 defects identified and fixed.

Reference: Technical Validation Report, Sections 1–5.

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2. Overall Testing Coverage

The validation covered the following major areas:

Testing Area	What Was Tested
Market Regimes	Bullish, bearish, sideways, volatile and uncertain markets
Strategy Selection	Directional, spread, neutral and defined-risk strategies
Market Conflicts	Stock vs sector, stock vs index and conflicting signals
Technical Indicators	RSI, ADX, momentum, trend and price structure
Breakouts	Confirmed, failed and low-edge breakouts
Gaps	Gap continuation, gap fade and unresolved gaps
Volatility	IV, VIX, volatility expansion and contraction
Events	Results, RBI/events and event timing
Expiry	0–1 DTE, expiry day and longer expiries
Session Timing	Market open, normal session and closing period

Liquidity	Volume, OI, spread and order-book depth
Option Chain	Missing strikes, duplicate contracts and incomplete chains
Greeks	Delta, IV and invalid/impossible Greeks
Data Quality	Stale, missing, corrupt and contradictory data
Risk	Maximum loss, stress loss and user risk limits
Trade Construction	Strike selection, payoff and breakevens
OI Analytics	OI buildup, walls, migration, PCR and max pain
Adversarial Testing	Deliberately incorrect or misleading inputs
Historical Testing	Walk-forward backtesting
Regression	Existing rules tested after changes

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3. Market Regime Testing

The Radar was tested against different overall market regimes to verify that strategy selection changes according to market conditions.

Case 3.1 — Strong Bullish Market

Scenario:
The underlying shows a strong upward trend with positive momentum and supporting market conditions.

What was tested:
Whether the Radar recognizes a clean bullish setup.

Expected behavior:
A directional bullish strategy can qualify when liquidity, risk and execution conditions also pass.

Example outcome:
`Qualified → Long Call`

Reference: RADAR-001.

Case 3.2 — Mildly Bullish Market

Scenario:
The stock is bullish, but the trend is not strong enough to justify an aggressive outright directional position.

What was tested:
Whether strategy selection adjusts to the strength of the signal.

Expected behavior:
A defined-risk bullish spread may be preferred over an outright long option.

Example strategy:
`Bull Call Spread`

Case 3.3 — Strong Bearish Market

Scenario:
The underlying has strong bearish structure and momentum.

What was tested:
Whether the Radar can identify bearish opportunities.

Expected behavior:
A bearish directional strategy can qualify when all other conditions pass.

Example outcome:
`Qualified → Long Put`

Reference: RADAR-009.

Case 3.4 — Sideways Market

Scenario:
The underlying remains within a defined range without a clear directional trend.

What was tested:
Whether the Radar avoids forcing a directional trade.

Expected behavior:
Neutral/range strategies can be considered when volatility and risk conditions support them.

Case 3.5 — Sideways + High Volatility

Scenario:
Price remains range-bound while volatility is elevated.

What was tested:
Whether the Radar combines price structure and volatility when selecting a strategy.

Expected behavior:
Defined-risk neutral strategies such as:

- Iron Condor
- Butterfly

may qualify when all required conditions pass.

References: RADAR-004, RADAR-020.

Case 3.6 — Sideways + Low Volatility

Scenario:
The market is sideways but volatility is not sufficiently attractive.

What was tested:
Whether the system avoids automatically selling premium merely because the market is range-bound.

Expected behavior:
`No Trade` when the required edge is not available.

Reference: RADAR-005.

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4. Directional Signal Testing

Case 4.1 — Overbought During Strong Uptrend

Scenario:
RSI is high/overbought while the underlying remains in a strong bullish trend.

What was tested:
Whether the Radar incorrectly treats an overbought reading as an automatic bearish signal.

Expected behavior:
The bullish trend can remain valid when other confirmation conditions pass.

Reference: RADAR-012.

Case 4.2 — Oversold During Strong Downtrend

Scenario:
RSI is oversold while the underlying remains in a strong bearish trend.

What was tested:
Whether oversold conditions automatically invalidate the bearish setup.

Expected behavior:
The bearish setup can remain valid when the broader structure supports it.

Reference: RADAR-011.

Case 4.3 — Counter-Trend Rally

Scenario:
A bearish stock temporarily rallies without sufficient evidence of a trend reversal.

What was tested:
Whether a temporary move is incorrectly interpreted as a new bullish trend.

Expected behavior:
No directional opportunity when confirmation is insufficient.

Reference: RADAR-010.

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5. Breakout Testing

Case 5.1 — Confirmed Breakout

Scenario:
Price breaks a resistance level with supporting confirmation.

What was tested:
Whether the Radar recognizes a genuine breakout rather than treating all price movements equally.

Expected behavior:
A bullish opportunity may qualify when the expected edge exceeds costs and risk requirements.

Case 5.2 — Fake Breakout

Scenario:
Price briefly moves above resistance but fails to maintain confirmation.

What was tested:
Whether the system avoids chasing an unreliable breakout.

Expected behavior:
`No Trade`.

Reference: RADAR-002.

Case 5.3 — Breakout With Insufficient Edge

Scenario:
A technically confirmed breakout exists, but the expected opportunity is too small after estimated costs.

What was tested:
Whether technical confirmation alone can trigger a trade.

Expected behavior:
`No Trade`.

Reference: EDGE-14.

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6. Breakdown Testing

Case 6.1 — Confirmed Breakdown

Scenario:
Price breaks important support with bearish confirmation.

What was tested:
Whether the Radar can recognize a bearish structural move.

Expected behavior:
A bearish strategy can qualify if risk, liquidity and cost conditions pass.

Case 6.2 — Failed Breakdown

Scenario:
Price temporarily falls below support and then recovers.

What was tested:
Whether the system avoids treating a failed breakdown as a valid bearish signal.

Expected behavior:
`No Trade` or neutral handling depending on the remaining market structure.

Reference: STR-01.

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7. Gap Testing

Gap scenarios were tested separately because gaps can materially change the market structure within a short period.

Case 7.1 — Gap-Down Continuation

Scenario:
The stock gaps downward and continues moving in the same bearish direction.

What was tested:
Whether the Radar recognizes continuation instead of assuming an automatic gap reversal.

Expected behavior:
Bearish opportunity may qualify.

Reference: GAP-01.

Case 7.2 — Gap Fully Recovered

Scenario:
The stock gaps down but subsequently recovers the gap.

What was tested:
Whether the system recognizes a reversal/range condition.

Expected behavior:
A neutral/range structure may be considered instead of blindly selecting a bearish trade.

Reference: GAP-02.

Case 7.3 — Unresolved Gap

Scenario:
A significant opening gap remains unresolved for the required period.

What was tested:
Whether the system waits rather than immediately generating a trade.

Expected behavior:
The setup can be blocked until the required condition is satisfied.

Reference: RADAR-015a.

Case 7.4 — Gap Fade

Scenario:
The initial gap begins to reverse and price moves back toward the previous reference level.

What was tested:
Whether the Radar distinguishes a gap-fade environment from a continuation environment.

Expected behavior:
Neutral/range structures may become relevant.

Reference: RADAR-015b.

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8. Market and Sector Conflict Testing

Case 8.1 — Bullish Stock + Bearish Sector

Scenario:
The stock is bullish, but its sector is strongly negative.

What was tested:
Whether sector-level information can downgrade a stock-level opportunity.

Expected behavior:
The setup can be moved to `WATCH` rather than automatically qualified.

Reference: DIR-02.

Case 8.2 — Bullish Stock + Mildly Soft Sector

Scenario:
The sector is slightly negative but not beyond the configured conflict threshold.

What was tested:
Whether the conflict logic is too aggressive.

Expected behavior:
The stock can still qualify if the other conditions pass.

Reference: DIR-02b.

Case 8.3 — Bearish Stock + Strongly Positive Sector

Scenario:
The stock is bearish while the sector is strongly positive.

What was tested:
Whether the Radar recognizes the opposite conflict.

Expected behavior:
Opportunity can be downgraded to `WATCH`.

Reference: DIR-02c.

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9. Broad Market Conflict Testing

Case 9.1 — Strong Bullish Stock + Strongly Bearish NIFTY

Scenario:
The individual stock is strongly bullish but the broader market is strongly bearish.

What was tested:
Whether a major market conflict is detected.

Expected behavior:
The opportunity should not automatically qualify.

Reference: DIR-06.

Case 9.2 — NIFTY With Mildly Bearish Conditions

Scenario:
NIFTY is only mildly bearish.

What was tested:
Whether mild market weakness is treated differently from a severe market conflict.

Expected behavior:
The stock opportunity is not automatically blocked solely because of mild index weakness.

Reference: DIR-04.

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10. Volatility Testing

Case 10.1 — High IV + Bullish Structure

Scenario:
The underlying is bullish but option volatility is expensive.

What was tested:
Whether the Radar considers the cost of buying volatility.

Expected behavior:
A spread may be preferred over an outright long option where the strategy rules support it.

Reference: RADAR-021.

Case 10.2 — Extreme IV

Scenario:
Implied volatility reaches an extreme level, such as approximately 250%.

What was tested:
Whether extreme volatility values can generate unreliable strategies.

Expected behavior:
`No Trade`.

Reference: EDGE-13.

Case 10.3 — VIX Spike / Panic Condition

Scenario:
The market experiences a significant volatility spike.

What was tested:
Whether the Radar detects panic conditions.

Expected behavior:
Trade generation is restricted during the configured panic state.

Reference: STR-05.

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11. Event Risk Testing

Case 11.1 — Results/Event Tomorrow

Scenario:
A stock has an otherwise attractive setup, but a major scheduled event is approaching.

What was tested:
Whether event risk overrides an otherwise attractive setup.

Expected behavior:
The trade is blocked when event rules require it.

Reference: RADAR-006.

Case 11.2 — Event on a Different Symbol

Scenario:
A high-impact event exists elsewhere in the market but is not directly relevant to the selected stock.

What was tested:
Whether the Radar incorrectly blocks unrelated opportunities.

Expected behavior:
The unrelated event should not automatically block the trade.

Reference: EVT-02.

Case 11.3 — Multiple Events on Same Day

Scenario:
Low-impact and high-impact events occur on the same day.

What was tested:
Whether the event engine correctly determines which event has greater significance.

Expected behavior:
The higher-impact event governs the decision.

Reference: EVT-05.

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12. Expiry Testing

Case 12.1 — Very Near Expiry

Scenario:
The selected contract has approximately one day or less remaining.

What was tested:
Whether the Radar avoids inappropriate long-premium trades when time decay and gamma risk become significant.

Expected behavior:
Outright long premium may be blocked.

Reference: TIME-01.

Case 12.2 — Expiry Day

Scenario:
The option expires on the current trading day.

What was tested:
Whether expiry-day conditions are handled separately from normal trading days.

Expected behavior:
Certain strategies are blocked because of increased premium/gamma sensitivity.

Reference: TIME-05.

Case 12.3 — Expiry Close Approaching

Scenario:
Only a short period remains before the market closes on expiry.

What was tested:
Whether the system stops generating opportunities too close to the session end.

Expected behavior:
`No Trade`.

Reference: EDGE-01.

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13. Market Session Timing

Case 13.1 — Before Market Open

Scenario:
A theoretically attractive setup is submitted before the configured trading session.

What was tested:
Whether the Radar respects market-session boundaries.

Expected behavior:
`Session Closed → No Trade`.

Reference: EDGE-03.

Case 13.2 — Normal Market Open

Scenario:
A valid setup appears after the required market warm-up period.

What was tested:
Whether valid opportunities can become eligible after the session begins.

Expected behavior:
The setup can qualify if all conditions pass.

Reference: TIME-03.

Case 13.3 — After Closing Cutoff

Scenario:
A setup arrives after the configured closing cutoff.

What was tested:
Whether the Radar prevents new opportunities after the allowed trading window.

Expected behavior:
`No Trade`.

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14. Liquidity Testing

Case 14.1 — Sufficient Volume and OI

Scenario:
The option has adequate trading activity and open interest.

What was tested:
Whether normal liquidity allows trade construction.

Expected behavior:
Liquidity gate passes.

Case 14.2 — Zero Volume/OI

Scenario:
An option has zero or insufficient trading activity.

What was tested:
Whether theoretical contracts can accidentally become executable suggestions.

Expected behavior:
`Liquidity Failure`.

Reference: EDGE-05.

Case 14.3 — Wide Bid/Ask

Scenario:
The option has a bid/ask spread above the permitted threshold.

What was tested:
Whether the Radar accounts for realistic execution costs.

Expected behavior:
`No Trade`.

Reference: RADAR-007.

Case 14.4 — Insufficient Order-Book Depth

Scenario:
The option technically has a quote but insufficient available quantity.

What was tested:
Whether a quote alone is considered sufficient evidence of executable liquidity.

Expected behavior:
Trade rejected.

Reference: EDGE-06.

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15. Option-Chain Integrity Testing

Case 15.1 — Missing Strikes

Scenario:
Some strikes are unavailable in the option chain.

What was tested:
Whether the strategy engine can construct a valid defined-risk trade only when the required contracts actually exist.

Expected behavior:
Unavailable legs are not fabricated.

Reference: DATA-08.

Case 15.2 — Duplicate Contracts

Scenario:
Duplicate contract records are supplied.

What was tested:
Whether duplicate data can corrupt strategy construction.

Expected behavior:
The structural validation remains safe.

Reference: DATA-07.

Case 15.3 — Only Calls Available

Scenario:
The option chain contains calls but puts are unavailable.

What was tested:
Whether the strategy engine adapts to the available contract universe.

Expected behavior:
Call-based opportunities can still be constructed where valid.

Reference: SHP-01.

Case 15.4 — Only Puts Available

Scenario:
Only put contracts are available.

What was tested:
Whether the system can construct valid bearish structures from available contracts.

Expected behavior:
Put-based opportunities can be considered.

Reference: SHP-02.

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16. Stale Quote Testing

Case 16.1 — Fresh Quotes

Scenario:
Option quotes are current and within the permitted age.

Expected behavior:
Quotes can participate in trade construction.

Case 16.2 — Stale Quotes

Scenario:
The option quote has not updated for the configured time period.

What was tested:
Whether stale pricing can incorrectly produce an executable opportunity.

Expected behavior:
Stale quotes are rejected.

Reference: DATA-04, SHP-03.

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17. Data-Quality Testing

Case 17.1 — High Data Quality

Scenario:
The data-quality score is above the required threshold.

Expected behavior:
The system can continue evaluating the opportunity.

Case 17.2 — Watch-Level Data Quality

Scenario:
Data quality deteriorates but remains above the hard rejection threshold.

Expected behavior:
Opportunity can be downgraded to `WATCH`.

Reference: DATA-02.

Case 17.3 — Low Data Quality

Scenario:
Data quality falls below the minimum acceptable level.

Expected behavior:
`No Trade`.

Reference: DATA-03.

Case 17.4 — Spot Price Missing/Invalid

Scenario:
Spot price becomes zero, NaN or otherwise invalid.

What was tested:
Whether invalid underlying prices can enter the strategy engine.

Expected behavior:
Data gate blocks the opportunity.

Reference: DATA-10.

Case 17.5 — Impossible OHLC Data

Scenario:
Market data contains impossible OHLC relationships.

What was tested:
Whether corrupted price data can create a false signal.

Expected behavior:
The decision remains safe and does not generate an invalid trade.

Reference: DATA-09.

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18. Feed-Conflict Testing

Case 18.1 — Spot and Futures Agree

Scenario:
Spot and futures data indicate the same market direction.

Expected behavior:
No conflict is generated.

Reference: FEED-01.

Case 18.2 — Spot and Futures Disagree

Scenario:
Spot and futures indicate materially different conditions.

What was tested:
Whether contradictory market feeds are detected.

Expected behavior:
The opportunity is treated as a data conflict rather than automatically traded.

Reference: DATA-01.

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19. Greeks Testing

Case 19.1 — Valid Greeks

Scenario:
Delta, IV and other required Greek inputs are within valid ranges.

Expected behavior:
Strategy construction proceeds normally.

Case 19.2 — Invalid/NaN Greeks

Scenario:
One or more Greek values are missing, NaN or mathematically invalid.

What was tested:
Whether invalid Greeks can corrupt payoff or strategy decisions.

Expected behavior:
Invalid values are handled safely.

Reference: EDGE-12 and defect validation.

Case 19.3 — Impossible Greek Values

Scenario:
Inputs contain mathematically impossible Greek values.

Expected behavior:
The system rejects or safely handles the invalid condition.

Reference: Defect validation in Section 8.

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20. Risk-Limit Testing

Case 20.1 — Trade Within User Risk Limit

Scenario:
The maximum loss remains within the configured per-trade risk limit.

Expected behavior:
Risk gate passes.

Case 20.2 — Maximum Loss Above User Limit

Scenario:
The proposed structure can lose more than the user's configured maximum risk.

Expected behavior:
`No Trade`.

Case 20.3 — Stress Loss Above Allowed Threshold

Scenario:
Normal maximum loss may appear acceptable, but stress testing produces excessive exposure.

Expected behavior:
Trade is rejected when stress loss exceeds the configured protection threshold.

The validation used a stress-loss rejection rule above 1.5× the user's per-trade limit.

Reference: Section 6, Capital Protection.

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21. Defined-Risk Strategy Testing

The Radar was tested to ensure suggested automated structures have clearly defined risk characteristics.

Strategies tested included structures such as:

- Long Call
- Long Put
- Bull Call Spread
- Bear Put Spread
- Bull Put Spread
- Bear Call Spread
- Iron Condor
- Butterfly
- Put Butterfly
- Calendar

- Diagonal
- Collar
- Protective Put
- Backspreads

The validation checked:

- Leg structure
- Strike ordering
- Maximum loss
- Maximum profit
- Breakeven
- Contract availability
- Liquidity
- Executable prices
- Payoff consistency

Reference: Sections 5–7.

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22. Trade Construction Validation

Every independently re-verified suggestion was checked for structural correctness.

Case 22.1 — Correct Strike Ordering

Test:
Verify that strikes are ordered correctly for the selected strategy.

Case 22.2 — Correct Contract Availability

Test:
Verify that every suggested leg exists in the actual option chain.

Case 22.3 — Executable Fill Prices

Test:
Verify that suggested prices are realistic relative to bid/ask prices.

Case 22.4 — Correct Maximum Loss

Test:
Independently recalculate the maximum possible loss.

Case 22.5 — Correct Breakeven

Test:
Independently recalculate breakeven levels.

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23. Deliberately Broken Trade Testing

The validator was intentionally given incorrect trade structures to determine whether it could identify errors.

The system successfully detected 8 deliberately injected trade defects.

Cases included:

1. Inverted strikes
2. Incorrect maximum loss
3. Incorrect breakeven
4. Fill price better than available bid
5. Illiquid option leg
6. Strike that was not listed
7. Short put positioned incorrectly relative to spot
8. Iron Condor shorts not correctly positioned around spot

Purpose:
This verifies that validation is not simply accepting whatever the strategy generator produces.

Reference: Section 5, Suggestion Correctness.

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24. OI Analytics Testing

Open-interest analytics were separately tested.

Cases included:

- OI buildup
- OI reduction
- OI walls
- OI migration
- Put/Call Ratio
- Max Pain
- Support/resistance derived from OI

These signals were validated as analytics/display information rather than automatically treated as trade-decision signals until sufficient out-of-sample validation exists.

Reference: Section 5.

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25. Threshold Boundary Testing

Boundary testing checks what happens immediately before and after an important rule threshold.

This is important because a system may work correctly at normal values but fail exactly at the decision boundary.

Examples tested included:

Rule	Passing Case	Rejecting Case
Panic threshold	-2.9%	-3.0%
VIX spike	-1.4	-1.5

Opening gap	0.9%	1.0%
Gap resolution	61 min	59 min
Market warm-up	09:30	09:29
Closing cutoff	15:14	15:15
Data quality	0.60	0.59
READY score	0.85	0.84
Minimum leg volume	100	99
Bid/ask spread	9.6%	10.6%
Sector conflict	-0.99%	-1.00%
NIFTY trend score	-0.39	-0.40

Purpose:
To verify that configured thresholds are actually enforced and that one small change in input produces the correct state transition.

Reference: Section 4, Threshold Boundary Testing.

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26. Configuration Testing

The Radar was tested with different administrative configurations.

Case 26.1 — Strategy Disabled

Scenario:
A strategy is disabled in configuration.

Expected behavior:
The Radar must not suggest that strategy.

Reference: CFG-01.

Case 26.2 — All Strategies Disabled

Scenario:
Every strategy is disabled.

Expected behavior:
`No Trade`.

Reference: CFG-02.

Case 26.3 — Unrealistic Risk/Reward Requirement

Scenario:
The minimum risk/reward configuration is set to an unrealistic level.

Expected behavior:
Trades failing the configured requirement are rejected.

Reference: CFG-03.

Case 26.4 — Extremely Strict Liquidity Requirement

Scenario:
Liquidity configuration is intentionally made too restrictive.

Expected behavior:
The system correctly rejects opportunities that cannot satisfy the requirement.

Reference: CFG-04.

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27. Black-Swan and Adversarial Testing

The system was also tested against abnormal market conditions.

Examples included:

- Market crash
- Extreme volatility
- Contradictory feeds
- Invalid spot price
- Corrupt OI
- Invalid VIX
- Impossible Greeks
- Stale quotes
- Missing contracts
- Extremely thin order books
- Extreme option IV

Purpose:
The objective was to determine whether the Radar fails safely rather than generating an apparently attractive trade from corrupted or abnormal inputs.

Reference: EDGE, DATA, FEED and adversarial scenario groups in Section 4.

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28. Multiple-Strategy Conflict Testing

Case 28.1 — More Than One Strategy Valid

Scenario:
Market conditions satisfy the requirements of multiple strategies simultaneously.

What was tested:
Whether the Radar can identify multiple valid alternatives instead of producing contradictory outputs.

Expected behavior:
Valid alternatives can be returned according to the strategy-selection rules.

Reference: RADAR-022.

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29. Insufficient Historical Data

Case 29.1 — New Listing / Limited History

Scenario:
A newly listed instrument has insufficient historical observations.

What was tested:
Whether the system generates signals despite insufficient history.

Expected behavior:
Opportunity is blocked until sufficient data exists.

Reference: RADAR-023, DATA-11.

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30. Walk-Forward Historical Validation

The Radar was also tested against historical data rather than relying only on synthetic scenarios.

The validation used rolling walk-forward testing across the historical period covered by the technical report.

The test universe included:

- 53 F&O underlyings
- Multiple strategy types
- Multiple market regimes
- Historical daily observations

- Transaction costs
- Slippage assumptions
- Separate training and validation periods

Approximately 119,176 modelled strategy trades were evaluated during the historical validation.

The purpose was to test whether the strategy-selection and trading rules remained consistent across different historical periods rather than only succeeding on a single selected period.

Reference: Section 7, Historical Backtest.

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31. NIFTY Strategy Testing

A separate NIFTY analysis was performed because recorded India VIX data was available for the index.

The analysis tested premium-selling defined-risk structures under historical conditions.

The results showed positive historical performance for some premium-selling strategies in the tested periods, but the available sample did not establish statistical significance at the required level.

The purpose of this test was therefore to determine whether the observed strategy behavior warranted further validation rather than treating the historical result as proof of a guaranteed edge.

Reference: Section 7, NIFTY Deep Dive.

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32. Defect Discovery and Fix Validation

Testing was also used to identify implementation defects.

The validation process identified 17 defects, including issues involving:

- Pricing-model forward bias
- Butterfly wing sizing
- Event/IV-crush handling
- Support/resistance targets
- Credit-spread clearance

- Invalid/NaN Greeks
- Broken spot data
- Broad-market conflicts
- Sector conflicts
- India VIX handling
- Corporate actions
- Transaction-cost assumptions
- Paper-trade settlement
- Mixed lot sizes
- Impossible Greeks
- Corrupt OI/VIX scoring
- Expired nearest-expiry contracts

Each identified issue was addressed and included in subsequent validation.

Reference: Section 8, Defects Found and Fixed.

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33. Regression Testing

After changes to the Radar logic, the existing scenario library was used to verify that previously working behavior remained intact.

Regression testing covered:

- Strategy selection
- Risk rules
- Market-state detection
- Data gates
- Liquidity rules
- Expiry rules
- Event rules
- Trade construction
- Payoff calculations
- Conflict handling
- Configuration rules

The objective was to ensure that fixing or changing one rule did not unintentionally break another part of the Radar.

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34. Web Application Testing

The web application was also tested separately from the backend logic.

The validation included:

- Opportunity display
- Strategy information
- Trade legs
- Payoff information
- Risk information
- Radar decision states
- Data presentation
- Frontend/backend integration

The validation report records 96 web application tests.

Reference: Section 3.

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35. Automated Test Results

The complete automated validation recorded:

Test Category	Result
Backend automated tests	3,138
Web application tests	96
Total automated tests	3,234
Automated failures	0

Market scenarios	607
Contract profiles	4
Scenario runs	2,352
Independently re-verified suggestions	1,097
Deliberately broken trades detected	8 / 8
Defects identified and fixed	17

Reference: Technical Validation Report, Sections 1–5 and 8.

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36. What These Cases Demonstrate

The test suite was designed to validate the Radar at multiple levels.

Market Understanding

The system was tested across:

- Bullish markets
- Bearish markets
- Sideways markets
- High-volatility markets
- Low-volatility markets
- Gap markets
- Breakouts
- Breakdowns
- Event-driven conditions

Decision Quality

The system was tested for both:

- Positive decisions — when conditions are sufficient for an opportunity.
- Negative decisions — when the correct behavior is to avoid generating a trade.

Risk Protection

The system was tested against:

- Excessive maximum loss
- Excessive stress loss
- Illiquidity
- Wide spreads
- Invalid contracts
- Expiry risk
- Poor data
- Extreme volatility

Trade Integrity

The system independently checked:

- Legs
- Strikes
- Prices
- Liquidity
- Maximum loss
- Maximum profit
- Breakevens
- Contract availability

Failure Safety

The system was intentionally exposed to:

- Corrupt data
- Missing data
- Contradictory data
- Invalid Greeks
- Extreme values
- Stale quotes
- Broken option chains
- Incorrect trades

The purpose was to ensure that invalid conditions result in rejection, watch states or safe handling rather than uncontrolled trade generation.
